QP CODE: 25901615

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR/ IMPROVEMENT/ REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER
Multi-Disciplinary Courses (MDC) - MG2MDCBOT100 - GARDENING AND LANDSCAPING
(2024 ADMISSION ONWARDS)

Duration: 1 Hour Max. Marks : 35

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A - Short Answer - Answer any 5 questions - Each question carries 1 marks

1	Name two toxins that Snake Plants remove from the air. [K] / [CO1, CO2]		
		formaldehyde, benzene, trichloroethylene, xylene, and toluene. ***(the question is outside the scope of syllabus, give full credit if attended)***	
2	What is the primary purpose of an herbal garden? [K] / [CO1, CO2]		
		To cultivate medicinal and aromatic plants for therapeutic, culinary, and educational use	
3	Name two amendments that can be added to soil to improve its fertility. [K] / [CO1, CO2]		
		Compost, manure, and vermicompost ***(or any other amendments***)	
4	What are the methods of artificial vegetative propagation? [U] / [CO4]		
		Cuttings, layering budding grafting and micropropagation	
5	State one reason why sprinkler irrigation is preferred over flood irrigation. [K] / [CO3]		
		Sprinkler irrigation is preferred over flood irrigation because it offers significantly higher water efficiency, reducing water wastage by eliminating the need for large, unlined conveyance channels and preventing excess surface runoff	
6	Contrast a hedge with a fence in garden design. [U] / [CO2]		
		Hedges and fences are both used to define property boundaries, provide privacy, and create structure in garden design, but they differ significantly in aesthetics, maintenance, and ecological impact. A hedge is a living, growing boundary of shrubs or trees, while a fence is a static, manufactured structure made of materials like wood, metal, or vinyl.	
7	What is the primary function of plants in landscape design? [K] / [CO2]		
		The primary function of plants in landscape design is to define, structure, and enhance outdoor spaces by combining aesthetic beauty with practical, functional roles, such as providing privacy, controlling erosion, and regulating microclimates.	
8	Which element of art in landscape design helps to control directional movement in the landscape? [U] / [CO2, CO5]		
		The primary element of art in landscape design used to control directional movement, guide the eye, and direct flow is Line	
			(1×5=5)

Part B - Short Essay - Answer any 4 questions
Each question carries 5 marks

9	Discuss the different types of shrubs used in landscaping.	[U] / [CO2]
	Landscaping shrubs are essential woody, multi-stemmed plants that provide structure, color, privacy, and habitat. Key types include evergreens for year-round color, deciduous shrubs for seasonal flowers, and flowering shrubs (like Hydrangeas) for focal points. Functions in Landscaping • Foundation Plants: Placed against houses to soften edges • Hedges/Screens: Used for privacy or to define property • .Specimen Plants: Chosen for distinct beauty to be planted alone Common Shrub Categories & Examples ➢ Evergreen Shrubs – Juniper , Thuja: Excellent for formal hedges, topiary, or edging due to dense growth and pruning tolerance. Camellia: Provides, glossy foliage and early spring or late winter flowers. ➢ Deciduous Shrubs (Seasonal Color & Texture) - :Hydrangea: Popular for large, showy flower clusters that change color based on soil acidity. ➢ Flowering & Accent Shrubs (Focal Points):Hibiscus: Brings tropical flair with large, colorful blooms that attract pollinators.	
10	Explain how landscaping enhances the aesthetic and functional value of outdoor space	[U] / [CO1
	Landscaping enhances outdoor spaces by blending aesthetic beauty—such as vibrant colors, varied textures, and curated focal points - with functional utility, including increased privacy, usable living areas, and energy efficiency. Enhancing Aesthetic Value (Visual Appeal) - ➢ Color and Texture: Using a mix of flowers, ornamental grasses, and trees creates dynamic, year-round interest. ➢ Structural Elements: Incorporating hardscapes like stone walkways, patios, and pergolas adds architectural structure and visual interest to natural green areas. ➢ Focal Points: Carefully placed trees or water features (fountains/ponds) draw the eye and establish a welcoming, high-end atmosphere. Enhancing Functional Value (Usability and Efficiency) ➢ :Outdoor Living Areas: Installing patios, outdoor kitchens, and fire pits creates functional spaces for cooking, entertaining, and relaxation. ➢ Privacy and Security: Strategically planted trees, hedges, or built-in fencing (living walls) provide natural screening from neighbors and increase security.	
11	What are the advantages of air layering ?	[U] / [CO4

		➢ High Success Rate for Difficult Plants: Effective for plants that rarely root from cuttings, such as mangos, lychees, and various ornamentals. ➢ Faster Maturity & Large Size: Produces large, established plants quickly, often allowing them to bear fruit or flowers much earlier than plants grown from seeds or cuttings. ➢ Low Risk to Parent Plant: The new plant remains attached to the parent for nourishment until its own root system develops, minimizing risk and stress to both. ➢ Genetic Fidelity: The new plant is a clone, ensuring it has identical characteristics, such as fruit quality, growth habit, and color ➢ .Simple & Low Infrastructure: Does not require greenhouse conditions, misting systems, or high humidity environments to be successful. ➢ Ideal for Rejuvenation: Perfect for propagating leggy or overgrown indoor plants like rubber plants or dieffenbachia, allowing you to create a new, compact plant while resetting the original. ➢ Works on Large Branches: Allows for the propagation of large, woody branches that would be impossible to root through traditional cuttings
12	Differentiate between simplicity and unity in landscape design? [An] / [CO2, CO5]	
		Simplicity in Landscape Design - Omitting unnecessary elements that do not add to the overall design, fostering a clean, organized, and uncluttered look. Application: ➢ Limiting Varieties: Using fewer species of plants, but planting them in larger groups. ➢ Restrained Palette: Restricting the color palette and materials (e.g., using only one type of stone for paths and walls).Functionality: Focusing on essential design elements to avoid confusing, "fussy" landscapes. Unity in Landscape Design The "thread" that ties all components together, ensuring that different elements harmonize to create a complete picture. Application: ➢ Repetition: Repeating the same plant types, colors, shapes, or materials throughout the landscape to create rhythm. ➢ Interconnection: Ensuring that different areas of the landscape (e.g., lawn and patio) flow together naturally ➢ Style Consistency: Matching the garden style to the architecture of the home.
13	Write a brief note on Bonsai. [U] / [CO2, CO3]	
		Bonsai is the Japanese art of growing and shaping miniature trees in containers to resemble full-sized, aged trees in nature. Key Aspects of Bonsai: ➢ Art and Nature: The goal is to create a realistic, miniature representation of nature while emphasizing artistic beauty. ➢ Techniques: Bonsai are kept small through careful pruning of branches and roots, wiring branches to guide their growth, and limiting soil volume. ➢ Maintenance: These trees are not genetically small; they require regular care, including consistent watering, fertilizing, and repotting. ➢ Species: While many woody-stemmed tree or shrub species can be used, popular choices include junipers, pines, elms, and maples. ➢ Aesthetics: A quality bonsai often portrays a sense of age and weathering, sometimes living for centuries, with design principles focusing on balance, harmony, and grace.
14	List any five implements used in gardening. [K] / [CO2]	

Common gardening implements:

1. Trowel: A small, handheld shovel used for digging small holes for planting, potting, and weeding.
2. Shovel: Used for digging, lifting, and moving large amounts of soil, compost, or gravel.
3. Rake: Used for leveling soil, gathering debris, and breaking up clumps of dirt.
4. Pruning Shears (Secateurs): Used for cutting and trimming small branches, stems, and pruning plants.
5. Hoe: A tool used to break up soil surfaces, remove weeds, and create furrows for seeds.

(or any five tools)

(5 X 4 = 20)

Part C – Essay - Answer any 1questions Each question carries 10 marks

15 Differentiate Approach grafting from Tongue grafting. U] / [CO4]

	Approach grafting	Tongue grafting
Definition:	A method where both the scion and the rootstock are grown on their own roots, brought close together, and joined.	A technique (often used in bench grafting) where a scion is joined to a rootstock, with interlocking cut "tongues" to provide security.
Rooting State	: Both plant parts remain attached to their respective parent plants until the union is fully established.	The scion is severed from its parent, while the rootstock is usually bare root or potted.
Process:	A small slice is cut from both stems at an intermediate point. These cut surfaces are matched, held together by tape, and only detached after the union matures.	Similar to a splice graft, but a second "tongue" slit is made on both components, allowing them to lock together.
Usage:	Used for plants that are difficult to graft by other methods, or to salvage a weak plant by grafting a new, potted seedling onto it.	Primarily for grafting dormant woody plants of similar, small thickness (roughly pencil-thick
Advantage:	High success rate because the scion receives water/nutrients from its own roots until fully healed	Provides superior structural strength and greater cambial contact, promoting faster healing.

16 Explain the process of identifying functional requirements in landscape design. How do functional requirements influence design decisions? [U] / [CO2]

<u>Identification of functional requirements in landscape design</u>

Identifying functional requirements in landscape design involves a systematic analysis of how the outdoor space will be used, maintained, and experienced by its users. The goal is to move beyond aesthetics to create a practical, usable space that meets the needs of family, pets, and visitors.

Activities for identification

1. Conduct a "Needs and Desires" Assessment
2. Perform a Site Inventory and Analysis
3. Create a Functional Diagram (Bubble Diagram)
4. Apply Key Design Principles
5. Document and Prioritize

<u>Influence of functional requirements in design decisions</u>

Functional requirements directly dictate design decisions by defining the specific tasks, features, and behaviors a system must perform. They influence architectural choices, data modeling, and user interface (UI) design to ensure the final product meets user needs and business objectives.

Major influencing areas

- ❖ Architectural Choices: Functional requirements dictate whether to use a microservices or monolithic architecture.
- ❖ Data Modeling & Storage: Requirements outlining how data is handled
- ❖ User Interface (UI) Design: Functional requirements determine user interactions

(liberal valuation)

(10×1=10)